

InsightEmb: Learning Action-Intent Embeddings for Agentic Insight Retrieval

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Abstract

Self-improving agents accumulate reusable insights from prior trajectories, making retrieval increasingly important for turning accumulated experience into actionable guidance. At each decision step, retrieving the right insight can help the agent progress toward its goal, a setting we refer to as **agentic insight retrieval**. However, existing retrieval methods primarily model semantic similarity, while overlooking whether a retrieved insight resolves the agent’s current decision bottleneck. We propose **InsightEmb**, a contrastive embedding framework that learns transferable progress-oriented retrieval geometry using *only* mathematical reasoning data. InsightEmb jointly learns to align concrete situations with abstract heuristic rules and to cluster reasoning trajectories with similar progress structures. We evaluate InsightEmb on dynamic agent tasks and static skill retrieval bench. Without any environment-specific training, InsightEmb improves over all these evaluations, surpassing the performance of existing reasoning embedding models. These results suggest that the geometry of state-insight matching can transfer across domains, enabling effective training from publicly available reasoning data without expensive environment-specific supervision.

1 Introduction

LLM-based agents in interactive environments, for examples, web navigation (Yao et al., 2022), embodied tasks (Shridhar et al., 2021), and tool use (Schick et al., 2023), must select actions under large action spaces where the optimal strategy depends on abstract reasoning rather than surface-level pattern matching. Providing agents with *insights*, abstract rules distilled from past experience (e.g., “check likely locations before exploring randomly”), has proven effective (Majumder et al., 2023; Wang et al., 2024; Zhao et al., 2024), but the agent must dynamically retrieve the most perti-

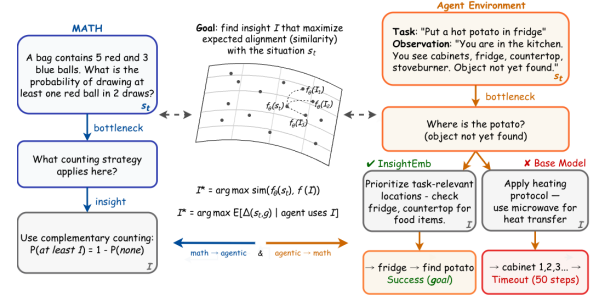


Figure 1: Overview of InsightEmb

nent insight at each step. We focus on this underspecified retrieval problem: the relevant insight is the one that is *operationally useful now*, given the agent’s current state, goal, and action history.

This dynamic retrieval poses a unique challenge that we call the *abstraction gap*: the query (the agent’s current observation and action history) and the target (an abstract insight) live at fundamentally different levels of abstraction. More importantly, relevance is not determined by topical similarity alone, but by the agent’s *next-step intent*: what bottleneck must be resolved before progress is possible? Standard embedding models match on surface features—keywords like “mug” or “stoveburner”—rather than functional relevance such as “the object has not been found, so the useful insight should guide search.” Consequently, off-the-shelf retrievers return topically related but procedurally premature insights, such as state-transformation rules before the agent has located the target object. It is illustrated in Figure 1.

Our key observation is that this *situation-to-insight* matching problem is not domain-specific. In mathematics, a probability query such as “ $P(\text{at least one red})$ in two draws” must be matched to complementary-counting heuristics with no topical overlap with the problem statement. In an embodied task, the same formalism applies: the agent must first resolve the *locate-object* bottle-

neck before heating or placement insights become useful. Both domains require inferring the latent bottleneck of a concrete situation and retrieving the abstract rule that enables the next effective step.

We propose **InsightEmb**, a contrastive training framework that exploits this domain-agnostic structure. InsightEmb trains an embedding model in two stages, *entirely on publicly available mathematical reasoning data*: (1) *Situation-to-insight matching* teaches the model to bridge the abstraction gap by mapping math problems and partial solution trajectories to their relevant heuristic rules; (2) *Situation-to-experience matching* teaches structural similarity recognition between reasoning trajectories, reinforcing the model’s ability to identify when different-looking situations require the same underlying strategy. At inference time, the trained model retrieves insights for LLM agents in ALF-World (Shridhar et al., 2021) and WebShop (Yao et al., 2022), and we use SRA-Bench as a static skill-retrieval diagnostic. These evaluations test dynamic state-conditioned retrieval and static skill applicability without any environment-specific training data.

Our contributions are threefold:

- We formulate agentic insight retrieval as goal-conditioned action-intent matching. Relevance is defined by whether an insight resolves the current bottleneck and enables progress, rather than by semantic similarity alone.
- We introduce InsightEmb, a cross-domain contrastive training framework for action-oriented retrieval geometry with empirical validation.
- Our empirically result across static and dynamic evaluations support the effectiveness of InsightEmb, and analyses show that the gains come from retrieving procedurally useful insights.

2 Related Work

Retrieval-Augmented LLM Agents. RAG (Lewis et al., 2020; Guu et al., 2020) has been extended to agentic settings for retrieving past experiences (Shinn et al., 2023; Majumder et al., 2023), tool documentation (Qin et al., 2024), and skill libraries (Wang et al., 2024). These systems retrieve *concrete* artifacts—specific examples, code, or API descriptions—where surface similarity is a reasonable proxy for relevance. In contrast, we retrieve *abstract insights* (general heuristic rules), where the query and target share no surface overlap, making standard retrieval

ineffective. Moreover, prior agentic RAG systems use off-the-shelf or in-domain-trained retrievers while neglecting the objective mismatching and embedding bottleneck in agentic tasks.

Experience-Based Agent Learning. Reflexion (Shinn et al., 2023) improves agents through verbal self-reflection; CLIN (Majumder et al., 2023) accumulates causal abstractions across episodes; Voyager (Wang et al., 2024) builds reusable skill libraries. ExpeL (Zhao et al., 2024) autonomously collects success and failure trajectories and distills cross-task insights into reusable natural-language rules without weight updates. These methods focus on *what* knowledge to accumulate. Our work addresses the orthogonal question of *how* to retrieve the right piece of accumulated knowledge at the right time, and is compatible with any knowledge-accumulation strategy.

Reasoning-Oriented Dense Retrieval. Dense passage retrieval (Karpukhin et al., 2020; Xiong et al., 2021) and instruction-tuned embedders (Wang et al., 2022; Su et al., 2023) improve lexical and semantic query–document matching, but still struggle when relevance requires multi-step reasoning rather than keyword overlap. The BRIGHT benchmark (Su et al., 2025) formalizes this gap with realistic queries across economics, mathematics, programming, and other domains where even strong MTEB leaders achieve low nDCG@10. Recent work therefore trains *reasoning-oriented* dense retrievers with contrastive objectives on reasoning-heavy query–document pairs. ReasonIR (Shao et al., 2025) fine-tunes by generating challenging queries and hard negatives from seed documents, aimed at general reasoning tasks. Llama-NV-Embed-Reasoning (NVIDIA, 2026) is a more compact bidirectional encoder trained with contrastive learning on BRIGHT-filtered corpora and public reasoning-embed datasets. It currently ranks as the strongest *open-source* embedding model on BRIGHT. BGE-Reasoner (Chen et al., 2025) pursues the same static matching objective with larger Qwen3-based embedders. These models optimize whether a passage *supports* a fixed query through semantic or logical entailment. These works define relevance by semantic and explanatory support. In contrast, agentic insight retrieval defines relevance by *next-step utility*: whether an insight resolves the current bottleneck and makes progress possible.

Agentic Benchmarks. ALFWorld (Shridhar et al., 2021) and WebShop (Yao et al., 2022) evaluate embodied and web-shopping agents in multi-step settings where the retrieval query is the evolving agent state rather than a fixed task description. AgentBench (Liu et al., 2024) extend from these environment and introduces a multi-dimensional benchmark LLMs with eight interactive environments to systematically evaluate Large Language Models (LLMs) as agents. SRA-Bench (Su et al., 2026) introduces skill-retrieval augmentation and evaluates static task-to-skill matching over a large corpus of gold and distractor skills across six task families; it complements the dynamic benchmarks because the query does not evolve with execution, isolating retrieval quality from online control.

3 Method

3.1 Problem Setting

An LLM agent operates in an interactive environment where, at each step t , it observes a state s_t (task description, action history, current observation) and must select an action a_t toward a goal g . The agent has access to an insight corpus $\mathcal{I} = \{I_1, \dots, I_N\}$, where each insight I_i is a natural-language description of abstract rules and strategies. We can view this task as *goal-conditioned abductive retrieval*: given the current state and goal, retrieve the insight that best identifies which intervention would make progress possible. Equivalently, it is an anticipatory retrieval problem whose relevance is defined by expected utility:

$$I^* = \arg \max_{I \in \mathcal{I}} \mathbb{E} [\Delta(s_t, g) \mid \text{agent uses } I], \quad (1)$$

where $\Delta(s_t, g)$ measures reduction of the gap between the current state and the goal state. This makes relevance goal-conditioned, state-dependent, and progress-oriented. In practice, we learn an embedding model f_θ that approximates this utility-based relevance in a shared vector space and retrieves by cosine similarity:

$$I^* = \arg \max_{I \in \mathcal{I}} \text{sim}(f_\theta(s_t), f_\theta(I)). \quad (2)$$

The core difficulty is the *abstraction gap*: s_t contains concrete details (specific objects, locations, actions) while I^* contains abstract rules (general strategies, heuristics). For agentic tasks, this gap is also temporal: the relevant insight is the one that

Component	Math	Agentic task
Situation	Problem statement	Task + observation
Trajectory	Solution steps	Action history
Insight	Solving heuristic	Execution strategy
Abstraction gap	“at least one red ball” $\leftrightarrow$ “complementary counting”	“You clean the butterknife” $\leftrightarrow$ “place”

Table 1: Structural analogy between math reasoning and agentic execution—our design claim for building action-intent embeddings from math-only contrastive training.

addresses the agent’s current bottleneck, not necessarily the one most topically related to the final task goal. For example, before the target object is found, a search insight is more useful than an insight about how to transform or place that object. We address this gap through a two-stage contrastive curriculum trained entirely on mathematical reasoning data. Our training design rests on a single claim: math heuristic retrieval and agentic insight retrieval instantiate the same action-intent matching problem (Table 1).

3.2 Structural Analogy for Action-Intent Retrieval

We propose that an agentic action-intent embedding can be learned from math-only data because both settings share the same retrieval structure: a concrete situation must be matched to the abstract rule that resolves the *current* bottleneck, not merely to topically related text.

Under this parallel, contrastive training on math should teach a domain-agnostic geometric property: embed a situation near the abstract rule that makes the next useful step apparent. The instruction prefix at inference (§3.5) frames queries in domain-neutral terms so this property can activate on ALFWorld and WebShop without environment-specific fine-tuning. The experiments below test whether this claim holds in practice.

3.3 Stage 1: Situation-to-Insight Matching

Stage 1 teaches the embedding model to bridge the abstraction gap by matching mathematical problems to their relevant heuristic rules.

Training data. Each training example is a triplet (q, I^+, I^-) where q is a query in one of three forms: (i) a raw math problem statement (*query-only*), (ii) a problem concatenated with its full chain-of-thought solution (*full trajectory*), or (iii) a problem

with a truncated solution (*partial trajectory*). I^+ is a positive insight set containing rules helpful for solving the problem; I^- is a negative (irrelevant) insight set.

Multi-granularity invariance. A critical design choice is that all three query forms for the same base problem share *identical* positive and negative insight sets. This forces the model to learn that the same insight should be retrievable regardless of how much progress has been made—a property directly needed for dynamic agentic retrieval, where the query grows with each step.

Training objective. We use the InfoNCE contrastive loss with in-batch negatives:

$$\mathcal{L}_1 = -\log \frac{\exp(\text{sim}(f_\theta(q), f_\theta(I^+))/\tau)}{\sum_{I \in \{I^+\} \cup \mathcal{N}} \exp(\text{sim}(f_\theta(q), f_\theta(I))/\tau)} \quad (3)$$

where $\tau = 0.01$ and $\mathcal{N}$ includes in-batch negatives (training group size 11). The data spans three math domains—counting & probability, number theory, and geometry—totaling 4,943 examples.

3.4 Stage 2: Situation-to-Experience Matching

Stage 1 teaches the model to match situations to *abstract rules*; Stage 2 complements this by teaching it to match situations to *structurally similar solved problems*. This reinforces the model’s ability to recognize when two different-looking situations require the same reasoning approach.

The training data consists of triplets (q, T^+, T^-) where q is a raw problem or a full trajectory, T^+ is a structurally similar solved problem, and T^- is a dissimilar one. The loss uses the same contrastive objective as Stage 1, with trajectories replacing insights. This stage uses 2,896 examples across the same three math domains.

3.5 Inference: Dynamic Insight Retrieval

At inference time, the trained embedding model is deployed for dynamic insight retrieval in target interactive environments *without any domain-specific fine-tuning*. At each step t : (1) the agent’s current state s_t is encoded with a task-specific instruction prefix;¹ (2) the top- k most similar insights are retrieved from the pre-encoded corpus; (3) the retrieved insights are prepended to the LLM’s prompt for action generation.

¹Instruction:

4 Experimental Setup

4.1 Training

We fine-tune Qwen3-Embedding-4B (Qwen Team, 2025) using DeepSpeed ZeRO-3 on 8 GPUs. Both stages use learning rate 1×10^{-5} with cosine scheduling, per-device batch size 8 with 2 gradient accumulation steps, temperature $\tau=0.01$, training group size 11, maximum passage length 1,536 tokens, 6 epochs, and warmup ratio 0.1. Stage 2 initializes from the Stage 1 checkpoint. Full hyperparameters are in Appendix H.

4.2 Evaluation Overview

Our goal is to test the design claim in Table 1: whether math-only training yields progress-oriented, action-intent retrieval in ALFWorld, WebShop, and SRA-Bench. We evaluate two complementary settings: *dynamic agentic retrieval* in ALFWorld and WebShop, where insights must help online decision making as the state changes, and *static skill retrieval* in SRA-Bench, where task descriptions are matched to useful skills without executing an agent trajectory. ALFWorld stresses embodied search and object manipulation, WebShop stresses product comparison and navigation, and SRA-Bench isolates task-to-skill retrieval quality. Following the evaluation setting of AgentBench (Liu et al., 2024), we report success rate for ALFWorld and average task score for WebShop.

4.3 Evaluation with Agentic Environment

4.3.1 ALFWorld

We evaluate on ALFWorld (Shridhar et al., 2021), a text-based embodied environment where agents perform household tasks. Each task requires a sequence of navigation and interaction actions, making ALFWorld a direct test of whether retrieval can select insights that unblock the agent’s current procedural phase. We use Qwen3-8B as the action-generating LLM with greedy decoding, a history window of 3 steps, maximum 50 steps per episode, and top-1 insight retrieval. We evaluate on the in-distribution split (140 games, seed 42).

4.3.2 WebShop

To validate dynamic retrieval beyond embodied tasks, we additionally evaluate on WebShop (Yao et al., 2022), where agents navigate product pages to purchase items matching natural-language instructions. This requires retrieving insights for attribute comparison, variant selection, loop recov-

Environment	Corpus	# Insights	Avg Words	Range
ALFWorld	BUNDLE	2,419	494	90–1,114
	ATOMIC	21,916	25	1–365
WebShop	BUNDLE	501	549	176–1,064
	ATOMIC	5,212	24	1–371

Table 2: Insight corpus statistics. BUNDLE = multi-rule insights; ATOMIC = individual rules extracted from bundles. Range = min–max words per insight.

ery, and multi-step purchase workflows. We use the same Qwen3-8B action-generating LLM with greedy decoding, a history window of 3 steps, maximum 50 steps per episode, top-1 dynamic insight retrieval, and 4 parallel environments. We evaluate on 500 test games (seed 42). We compare the same embedding model variants: Base, InsightEmb.

4.3.3 Insight corpora

We evaluate two insight corpus granularities per environment to disentangle the effect of the embedding model from the insight content. BUNDLE insights are multi-rule summaries, each containing several rules with full chain-of-thought reasoning. ATOMIC splits each bundle into individual rules with chain-of-thought removed. Table 2 summarises the corpus statistics for both environments.

4.4 SRA-Bench Retrieval Evaluation

To isolate retrieval quality from downstream action generation, we additionally evaluate on SRA-Bench, a static skill-retrieval benchmark spanning theorem proving, logical reasoning, tool use, contest math, medical calculation, and code-generation tasks. Each query is formed from the task information, and each candidate is represented by the full skill content, including the skill name, description, and body. The insight corpus for SRA-Bench is the 636 manually constructed gold skills plus the 26,262 web-collected distractor skills. We compare the original Qwen3-Embedding-4B checkpoint (Base) with our cross-domain trained checkpoint (InsightEmb) using recall at k ($R@k$) and normalized discounted cumulative gain at k ($N@k$) for $k \in \{1, 3, 5, 7, 10\}$. Following the benchmark setting, we report macro averages across datasets so that each task family contributes equally regardless of its number of examples.

5 Results

5.1 ALFWorld Results

We first evaluate whether retrieved insights improve online agent execution in ALFWorld, where

Insight	Embedding	ALFWorld	WebShop
{}	—	52.86	31.03
BUNDLE	Base	54.29	18.42
BUNDLE	InsightEmb	60.71	31.74
ATOMIC	Base	55.00	28.25
ATOMIC	InsightEmb	59.29	32.05

Table 3: Dynamic agent evaluation on ALFWorld and WebShop.

the query changes after every observation and action, and on WebShop (§5.2), where the agent must search product pages and complete purchases. Table 3 reports both metrics under the same retrieval protocol (top-1 dynamic insight retrieval; Qwen3-8B action model).

InsightEmb transfers to embodied control. All InsightEmb variants outperform the Base embedding model on in-distribution success rate. With both BUNDLE and ATOMIC insights, InsightEmb achieves about a 5-point improvement over Base. Since InsightEmb is trained exclusively on mathematical reasoning data, these gains are entirely from cross-domain transfer. Comparison with an ALFWorld-trained in-domain retriever is deferred to §6.3.

Per-task-type breakdown. Appendix Table 12 breaks down in-distribution success rates by ALFWorld task type. InsightEmb leads on 4 of 6 types, with the largest gains on *clean* and *find_two*, indicating a general structural-matching gain rather than a single-task shortcut.

Mechanistic analysis. We further analyze *why* retrieval helps on ALFWorld: multi-granularity query invariance (Appendix F.2), a qualitative hot-potato example of bottleneck-matched retrieval (Appendix F.3), rule-level topical vs. procedural matching on the ATOMIC corpus (Appendix F.4), and phase-aware step-conditioned dynamics (Appendix I).

5.2 WebShop Results

WebShop results are summarized in Table 3 (right column). We report average task score (0–1), which reflects partial credit for attribute matching under the AgentBench evaluation protocol.

InsightEmb transfers to web shopping. All InsightEmb variants outperform the Base embedding model, yielding a large performance gains over Base, e.g., a 75% relative gain with BUNDLE insights. This confirms that the structural matching capability learned from math transfers to a second, structurally distinct interactive environment.

Qualitative patterns. InsightEmb reduces zero-score games from 309 to 258 and increases perfect-score games from 9 to 13, indicating that it helps the agent complete tasks more successfully. Manual inspection of the divergent games reveals three recurring patterns (detailed examples in Appendix J):

1. **Variant selection awareness.** InsightEmb’s insights guide the agent to explicitly select product variants (color, size) before purchasing, while Base frequently skips this step, resulting in partial scores instead of perfect scores.
2. **Loop prevention.** Base gets stuck in search–browse–back loops for 21–50 steps; InsightEmb’s insights about session management and error recovery help the agent break out of unproductive cycles.
3. **Procedural sequencing.** InsightEmb retrieves insights that encode a sequential workflow (search → verify → select variants → buy), while Base retrieves topically relevant but procedurally vague insights.

These patterns mirror the ALFWorld finding (Appendix F.4): InsightEmb performs *procedural matching* rather than *topical matching*, retrieving insights that address the agent’s current bottleneck.

5.3 Static Skill Retrieval on SRA-Bench

We next evaluate static skill retrieval on SRA-Bench to isolate retrieval quality from downstream action generation. Tables 4 and 5 report macro-averaged and per-task-family retrieval performance, testing whether the learned geometry retrieves procedural skills that are useful for a task description even before an agent trajectory unfolds. InsightEmb improves over Base on every cutoff, with small gains at R@1 and substantially larger gains at higher cutoffs (+8.90 R@10 and +5.78 N@10).

This pattern indicates that cross-domain contrastive training primarily improves the ranking of relevant skills beyond the first position, which is especially useful when an agent can inspect or condition on multiple retrieved skills.

The only clear negative outlier is MedCalcBench at small cutoffs (Table 5). A plausible explanation is that medical calculation skills are highly template-like, lexically specialized, and domain-specific: correct retrieval often depends on exact medical score names, disease terms, biomarker or variable names, units, and other clinical terminology. The Base embedding model appears to preserve these fine-grained biomedical and entity-level cues, whereas InsightEmb’s cross-domain training encourages more abstract structural matching and may therefore downweight or smooth over such terminology at rank 1. This interpretation is also consistent with the original SRA-Bench results, where MedCalcBench already achieves about 90% R@1 and over 90% R@10 after reranking the BM25 top-50 candidates with different reranker models, indicating that lexical candidate generation is less of a bottleneck for this task. In contrast, task families such as theorem proving, tool use, contest math, and code generation benefit from abstraction-aware ranking because the query and useful skill can differ substantially in wording while sharing a reasoning or procedural structure. This distinction is important in practical retrieval-augmented skill selection pipelines: an LLM reranker can only inspect a limited candidate set, such as the BM25 top-50 skills, due to context-window constraints. Thus, task families for which lexical retrieval fails to place the correct skill within this window constitute the more important bottleneck. From this perspective, the strong results on LogicBench and CHAMP are especially encouraging, because they require more semantic or reasoning-oriented matching between task descriptions and skill content, where surface lexical overlap is weak. InsightEmb therefore improves the recall of relevant skills before reranking, expanding the effective coverage of downstream LLM-based selection.

6 Analysis and Comparisons

The results above establish that math-trained InsightEmb improves dynamic agent execution and static skill retrieval over the Base embedder. This section asks *why*: we compare reasoning-oriented embedders and alternative insight generators, val-

Embedding	R@1	R@3	R@5	R@7	R@10	N@1	N@3	N@5	N@7	N@10
Base	30.33	40.84	46.40	49.83	54.13	33.82	37.62	39.96	41.23	42.64
InsightEmb	31.37	46.39	54.25	58.56	63.03	36.29	42.00	45.29	46.92	48.42
Δ	+1.04	+5.55	+7.85	+8.73	+8.90	+2.47	+4.38	+5.33	+5.69	+5.78

Table 4: SRA-Bench macro-average retrieval results across six task families. Queries use task information and candidates use the full skill content. $R@k$ is recall at k and $N@k$ is normalized discounted cumulative gain at k . Table 5 breaks down per-task-family results at @1 and @10; the macro-average is computed by averaging those task families equally.

Task family	n	Base				InsightEmb			
		R@1	N@1	R@10	N@10	R@1	N@1	R@10	N@10
TheoremQA	747	48.19	48.19	74.43	60.63	58.10	58.10	86.61	72.37
LogicBench	760	1.58	1.58	40.39	19.53	6.32	6.32	39.34	20.18
ToolQA	1,430	16.22	16.22	44.83	28.84	29.65	29.65	62.45	45.69
CHAMP	223	13.86	19.73	38.34	26.97	19.06	29.15	52.58	38.48
MedCalcBench	1,100	92.73	92.73	98.18	95.80	63.27	63.27	98.18	81.52
BigCodeBench	1,140	9.42	24.47	28.62	24.09	11.85	31.23	39.04	32.27

Table 5: Per-task SRA-Bench retrieval results at @1 and @10. Queries use task information and candidates use full skill content. Bold marks the better value within each task family; ties are bolded for both embeddings.

idate in-domain fine-tuning against cross-domain training, then analyze whether the learned geometry matches our structural-analogy claim.

6.1 Scaling with Comparison to Reasoning-Oriented Retrievers

A natural question is whether reasoning-oriented retrievers trained for static, query–document matching can serve as drop-in replacements for InsightEmb in agentic settings. Using the same ALFWorld and WebShop agent protocol as Table 3, we compare Base, InsightEmb, and two reasoning-oriented embedding baselines: (i) **Llama-NV-Embed-Reasoning-3B** (NVIDIA, 2026) (3.2B; Llama-3.2-3B), the top open-source retriever on BRIGHT; and (ii) **ReasonIR-8B** (Shao et al., 2025) (8B; Llama-3.1-8B), trained with contrastive learning on synthetic reasoning-intensive pairs—the setting closest to ours among static retrievers. Both differ from InsightEmb (4B; Qwen3-Embedding-4B), which is trained only on math data. Beyond the top-1 results in Table 3, we sweep retrieval budgets $k \in \{1, 3, 5\}$ for all four embedders on both environments and both insight corpora (BUNDLE and ATOMIC). Figure 2 plots ALFWorld success rate (left) and WebShop average task score (right) as a function of k , using the arithmetic mean of BUNDLE and ATOMIC at each k ; per-corpus breakdowns are in Appendix A.

On ALFWorld, InsightEmb peaks at $k=5$ on ATOMIC (67.86%) and remains competitive with

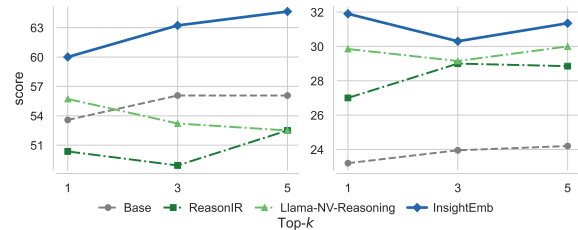


Figure 2: Top- k scaling for reasoning-oriented retrievers. **Left:** ALFWorld. **Right:** WebShop. Curves show the arithmetic mean of BUNDLE and ATOMIC retrieval at each $k \in \{1, 3, 5\}$ for Base, ReasonIR, Llama-NV-Reasoning, and InsightEmb. Per-corpus results are in Appendix A.

Llama-NV-Reasoning on BUNDLE at $k=1$, while reasoning-oriented retrievers do not consistently improve as k increases. On WebShop, ReasonIR and Llama-NV-Reasoning can surpass InsightEmb under BUNDLE insights at larger k , but InsightEmb retains the strongest top-1 ATOMIC score (0.321). These patterns reinforce that strong performance on static reasoning retrieval does not automatically translate to dynamic agentic insight retrieval. The gap reflects a distinction between two retrieval regimes:

- **Static abductive retrieval:** given a fixed query, retrieve content whose relevance must be reasoned about ex post. The query and candidate are fixed; relevance is judged by explanatory or logical support.
- **Dynamic action-intent retrieval** (agentic in-

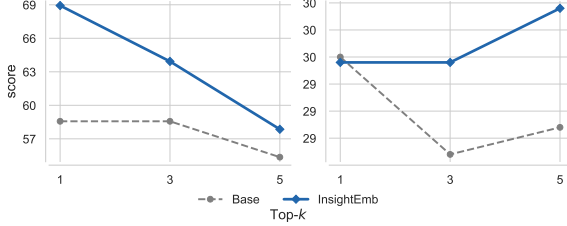


Figure 3: Robustness check with GPT-5.2-generated insight corpora. **Left:** ALFWorld. **Right:** WebShop. Curves average BUNDLE and ATOMIC at each k for Base and InsightEmb. Per-corpus results are in Appendix B.

sight retrieval): given an evolving agent state, retrieve the insight that resolves the *current procedural bottleneck*. Relevance changes with every action and depends on what the agent needs to do *next*.

Reasoning-oriented retrievers are tuned for the first regime—when a passage *explains* or *logically supports* a query. InsightEmb targets the second: when an abstract rule *enables progress* from a concrete situation. This distinction clarifies why cross-domain math training can outperform retrievers built for static reasoning benchmarks in embodied and interactive settings.

6.2 Robustness to Alternative Insight Generators

All main results in §5 use insight corpora distilled from agent trajectories with DeepSeek-R1. To test whether our gains depend on that particular generator, we repeat the dynamic-agent evaluation with corpora produced by GPT-5.2 instead, while keeping the action model (Qwen3-8B) and the same top- k retrieval protocol as above. Corpus statistics are in Appendix B.1; per-corpus top- k curves are in Appendix B.

Figure 3 plots the arithmetic mean of BUNDLE and ATOMIC at each $k \in \{1, 3, 5\}$. Results show that for both ALFWorld and WebShop, InsightEmb consistently outperforms Base, supporting that improvement persists under alternative insight generators.

6.3 Validating the Structural-Analogy Claim

Math-to-agentic transfer. Table 1 predicts *bidirectional* transfer: math training should improve agentic retrieval (tested in §5), and with this hypothesis, embodied training should improve math retrieval also. We validate both directions

Corpus	Embedding	ALFWorld
BUNDLE	Base	54.29
	ALF (In-domain)	57.86
	InsightEmb	60.71
ATOMIC	Base	55.00
	ALF (In-domain)	57.86
	InsightEmb	59.29

Table 6: ALFWorld in-distribution success rate (%) for an ALFWorld-trained in-domain retriever (BUNDLE only) vs. Base and math-trained InsightEmb (top-1 retrieval; Qwen3-8B agent). The in-domain model is trained and evaluated on the same environment.

and compare against the natural alternative of in-environment fine-tuning.

Finetuning With ALFWorld A standard alternative to cross-domain training is to fine-tune the embedder on ALFWorld situation-to-insight pairs. We train this *In-domain* retriever on the same backbone (Qwen3-Embedding-4B), using the same agent stack, 140 in-distribution games, and top-1 dynamic retrieval as §4. Table 6 contrasts it with Base and InsightEmb. InsightEmb exceeds the in-domain retriever on both BUNDLE and ATOMIC despite using no ALFWorld training data, suggesting that structural diversity in math supervision matters more than task-matched ALFWorld fine-tuning; we attribute this gap primarily to richer cross-domain abstraction diversity in math training (Appendix D).

Agentic-to-math transfer. For the reverse direction, we evaluate all model variants on *math insight retrieval*—retrieving the correct heuristic rule for a held-out math problem. Table 7 presents accuracy on four math domains. Three domains (number theory, geometry, and counting & probability) overlap with the training distribution (same domains, different problems from the test split with a separate insight corpus), while algebra is entirely out-of-distribution and never appears in training. InsightEmb consistently improves over Base on the held-out math test set and the unseen math domain. This confirms that the contrastive curriculum does not sacrifice in-domain math retrieval while learning the claimed geometry. More importantly, the ALFWorld-trained model (ALF) also improves over Base on math insight retrieval across all four domains. This reverse-direction gain supports Table 1: the geometry is shared, not a one-way math-to-agent shortcut.

Task	Base	InsightEmb	ALF
Number Theory*	77.22	79.81	78.33
Geometry*	54.07	59.29	57.41
Counting & Probability*	74.47	77.43	75.53
Algebra	86.69	88.21	87.81

Table 7: Math *insight* retrieval (R@1, %). Base = Qwen3-Embedding-4B; InsightEmb = Stage 1+2; ALF = ALFWorld-trained model. *In-domain (same domains as training, held-out test split with separate insight pool).

7 Discussion

Implications for agentic RAG. Our results suggest a practical paradigm for retrieval in data-scarce agentic settings: rather than collecting expensive in-domain retrieval training data, practitioners can train on data-rich reasoning domains (such as math) and transfer the learned action-intent matching capability. The key requirement is that the training data exhibits the same *situation* $\rightarrow$ *bottleneck* $\rightarrow$ *abstract rule* structure as the target domain. This is broadly satisfied by domains where concrete problems are solved by first identifying the latent constraint that blocks progress and then applying the corresponding abstract principle. Thus, the goal of an agentic retriever is not simply to retrieve text that describes the same objects or topic, but to retrieve the insight that makes the agent’s next useful action apparent.

8 Conclusion

We introduced InsightEmb, a contrastive training framework that learns to retrieve abstract reasoning insights for LLM agents by training exclusively on mathematical reasoning data. Our key finding is that agentic insight retrieval is a goal-conditioned action-intent matching problem: the model must connect a concrete state to the abstract rule that resolves the agent’s current bottleneck and enables progress toward the goal. We evaluate this capability in both dynamic and static settings: ALFWorld and WebShop test whether retrieval improves step-by-step agent execution, while SRA-Bench tests whether the same geometry retrieves useful skills from static task descriptions. InsightEmb improves ALFWorld success rates over the base embedding model and over an in-domain-trained retriever, demonstrates significant gains on WebShop, and improves SRA-Bench skill retrieval. These results point toward a general principle: when retrieval requires inferring what a solver or agent should do next, training on any domain that exhibits the same

situation-to-bottleneck-to-rule structure can yield effective cross-domain transfer.

Limitations. Our dynamic evaluation covers two target environments (ALFWorld and WebShop) with two action-generating LLMs (Qwen3-8B and GPT-5.2) and multiple retrieval budgets ($k \in \{1, 3, 5\}$), while SRA-Bench provides a static skill-retrieval diagnostic. Validating the approach on additional dynamic environments and with a wider range of LLMs would further strengthen the generality claim. Although SRA-Bench isolates static retrieval quality, dynamic agentic retrieval still lacks gold labels for which insight is optimal at each state; downstream success therefore remains an important but indirect measure of retrieval utility.

References

- Jianlyu Chen, Junwei Lan, Chaofan Li, Defu Lian, and Zheng Liu. 2025. [ReasonEmbed: Enhanced text embeddings for reasoning-intensive document retrieval](#). *arXiv preprint arXiv:2510.08252*.
- Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, and 1 others. 2024. [AgentBench: Evaluating LLMs as agents](#). In *International Conference on Learning Representations*.
- Bodhisattwa Prasad Majumder, Bhavana Dalvi Mishra, Peter Jansen, Oyvind Tafjord, Niket Tandon, Li Zhang, Chris Callison-Burch, and Peter Clark. 2023. CLIN: A continually learning language agent for rapid task adaptation and generalization. *arXiv preprint arXiv:2310.10134*.
- NVIDIA. 2026. [Llama-NV-Embed-Reasoning-3B](#). Hugging Face model card. 3.2B-parameter reasoning embedding model built on Llama-3.2-3B; top-ranked open-source retriever on BRIGHT.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, and 1 others. 2024. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *Proceedings of the Twelfth International Conference on Learning Representations*.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*.

Rulin Shao, Rui Qiao, Varsha Kishore, Niklas Muennighoff, Xi Victoria Lin, Daniela Rus, Bryan Kian Hsiang Low, Sewon Min, Wen-tau Yih, Pang Wei Koh, and Luke Zettlemoyer. 2025. [ReasonIR: Training retrievers for reasoning tasks](#). *arXiv preprint arXiv:2504.20595*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2021. ALFWorld: Aligning text and embodied environments for interactive learning. In *Proceedings of the Ninth International Conference on Learning Representations*.

Hongjin Su, Weijia Shi, Jungo Kasai, Yizhong Wang, Yushi Hu, Mari Ostendorf, Wen-tau Yih, Noah A. Smith, Luke Zettlemoyer, and Tao Yu. 2023. One embedder, any task: Instruction-finetuned text embeddings. In *Findings of the Association for Computational Linguistics: ACL 2023*.

Hongjin Su, Howard Yen, Mengzhou Xia, Weijia Shi, Niklas Muennighoff, Varsha Kishore, Luke Zettlemoyer, Wen-tau Yih, Danqi Chen, and Tao Yu. 2025. [BRIGHT: A realistic and challenging benchmark for reasoning-intensive retrieval](#). In *International Conference on Learning Representations*.

Weihsang Su, Jianming Long, Qingyao Ai, Yichen Tang, Changyue Wang, Yiteng Tu, and Yiqun Liu. 2026. [Skill retrieval augmentation for agentic AI](#). *arXiv preprint arXiv:2604.24594*.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.

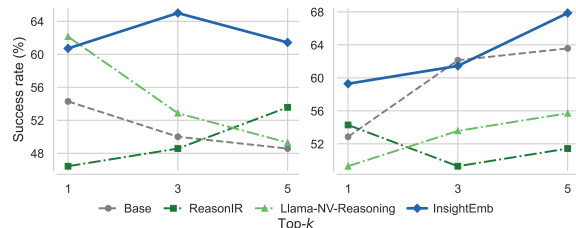


Figure 4: ALFWorld in-distribution success rate (%) vs. retrieval budget $k \in \{1, 3, 5\}$ for Base, ReasonIR, Llama-NV-Reasoning, and InsightEmb with BUNDLE and ATOMIC insight corpora.

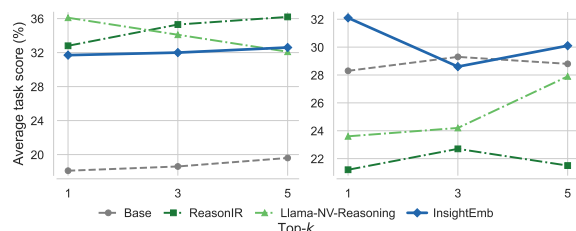


Figure 5: WebShop average task score at each k for the same embedding models and corpora (y -axis = AgentBench/WebShop score $\times 100$, e.g. 32.1 denotes 0.321).

Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul N. Bennett, Junaid Ahmed, and Arnold Overwijk. 2021. Approximate nearest neighbor negative contrastive learning for dense text retrieval. In *Proceedings of the Ninth International Conference on Learning Representations*.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. WebShop: Towards scalable real-world web interaction with grounded language agents. In *Advances in Neural Information Processing Systems*.

Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. [ExpEL: LLM agents are experiential learners](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19632–19642.

A Reasoning-Oriented Retriever Top- k Details

Figure 2 in the main text averages BUNDLE and ATOMIC at each k . Figures 4 and 5 show the per-corpus breakdown.

B Alternative Insight-Generator Robustness

B.1 Corpus Statistics

Table 8 summarizes GPT-5.2-generated insight corpora used in the robustness check (§6.2). BUNDLE

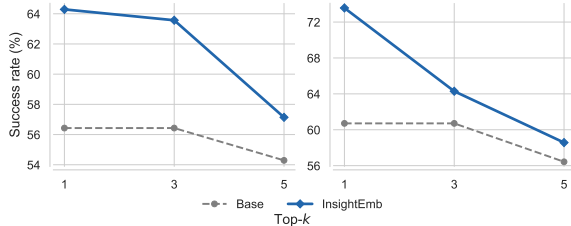


Figure 6: ALFWorld success rate (%) vs. $k \in \{1, 3, 5\}$ with GPT-5.2-generated insights.

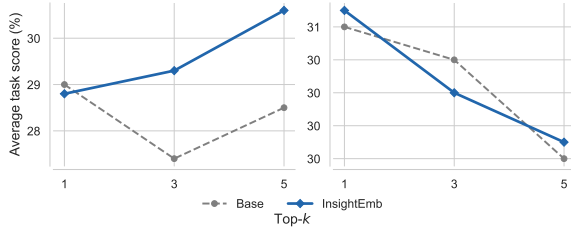


Figure 7: WebShop average task score vs. $k \in \{1, 3, 5\}$ with GPT-5.2-generated insights.

files contain one multi-rule summary per evolution chromosome ($n=700$ for each environment); ATOMIC files expand each bundle into individual rules (chain-of-thought removed). Compared with the DeepSeek-R1 corpora in Table 2, GPT-5.2 bundles are shorter on average while atomic rules are similar in length.

B.2 Top- k Scaling Details

Figure 3 in the main text averages BUNDLE and ATOMIC at each k . Figures 6 and 7 show the per-corpus breakdown (Qwen3-8B agent; same protocol as Table 3).

C Training Data Statistics

Stage	Total	Query	Traj
Stage 1	4,943	~1,650	~3,293
Stage 2	2,896	~2,329	~567

Table 9: Training data statistics. “Query” = query-only samples, “Traj” = full + partial trajectory samples.

Math Domain	Stage 1	Stage 2
Counting & Probability	1,650	966
Number Theory	1,650	966
Geometry	1,643	964
Total	4,943	2,896

Table 10: Task distribution in training data.

D Why Math Training Outperforms In-Domain Fine-Tuning

As shown in §6.3 (Table 6), the ALFWorld-trained retriever underperforms math-trained InsightEmb despite task-matched supervision. Two factors explain this gap. First, *data diversity*: math training covers three structurally distinct reasoning domains, each with its own abstraction patterns, while ALFWorld training data is limited to a single environment with repetitive task structures. Second, *abstraction range*: mathematical insights span from concrete formulas to high-level meta-strategies, providing a richer gradient of abstraction levels for the contrastive loss to exploit. These factors suggest that the quality of contrastive training for insight retrieval depends more on the *structural diversity* of the training domain than on its topical proximity to the target domain.

E Episode-Level Agent Statistics

Table 11 reports episode-level diagnostics for Qwen3-8B with top-1 dynamic insight retrieval on BUNDLE summary insights. **Wins**: ALFWorld success (won); WebShop perfect task score (task_score = 1). **Zero-score**: ALFWorld failures; WebShop task_score = 0. **Unique insights**: number of distinct retrieved insight blocks in prompts within an episode (deduplicated text blocks across steps).

F ALFWorld Per-Task and Mechanistic Analysis

F.1 Per-Task-Type Breakdown

F.2 Multi-Granularity Query Invariance

Our Stage 1 design enforces that query-only, partial-trajectory, and full-trajectory versions of the same problem all retrieve the same insight. This directly maps to the agentic setting: at step 0 the agent has only the task description (analogous to query-only), at mid-game it has partial action history (partial trajectory), and at late steps it has extensive history (full trajectory). The multi-granularity training ensures that the correct insight is retrievable at every stage of task execution, not just at the beginning.

F.3 Qualitative Example: Insight-Guided Search

Figure 8 illustrates how InsightEmb’s retrieved insights lead to better action selection on a represen-

Environment	Corpus	# Insights	Avg Words	Range
ALFWorld	BUNDLE	700	314	178–500
	ATOMIC	8,410	25	11–58
WebShop	BUNDLE	700	358	198–532
	ATOMIC	10,162	23	4–54

Table 8: GPT-5.2 insight corpus statistics. Range = min–max words per non-empty insight; WebShop has 10 empty bundle summaries (word count 0). Atomic = single rules split from bundles.

Metric	ALFWorld		WebShop	
	Base	InsightEmb	Base	InsightEmb
# games	140	140	500	500
# wins	76	85	5	8
# zero-score	64	55	309	258
Avg. task score	54.3%	60.7%	0.184	0.317
Avg. steps	30.3	28.2	40.8	33.9
Median steps	30	24	50	50
Avg. unique insights / game	1.82	5.79	1.85	5.98
Games with >1 unique	85	134	295	487
Max unique insights	4	22	4	18

Table 11: Episode-level statistics (Qwen3-8B, top-1 dynamic retrieval, BUNDLE insights). **Steps/Won/Lost/Med.** are action counts per episode (all / won only / lost only / median). **Uniq.** is mean unique insight blocks per episode; **>1** counts episodes with more than one distinct block; **Max** is the episode maximum. WebShop Base was evaluated on 432 test games in this run. ALFWorld lost episodes typically reach the 50-step cap.

Task Type	Base	Ours
clean (22)	27	41
cool (26)	62	69
heat (19)	42	53
put (48)	71	71
examine (17)	71	53
find_two (8)	12	38
Overall	55.0	59.3

Table 12: ALFWorld success rates by task type (top-1 retrieval, Qwen3-8B). Base = ATOMIC/Base; Ours = ATOMIC/InsightEmb. Numbers in parentheses indicate total games per type.

tative task (“put a hot potato in fridge”).

The base model’s insight is topically relevant but strategically vacuous—it does not indicate *where* to find the potato. InsightEmb retrieves a structurally matched search-priority rule, directly determining the agent’s first action. This pattern—generic vs. structurally specific retrieval—recurs across the 16 games that InsightEmb wins uniquely under top-1 retrieval (see also §G).

F.4 Topical vs. Procedural Retrieval: Rule-Level Analysis

Because each ATOMIC insight is a single rule, we can examine which *type* of rule each embedding model prioritizes for the same query. We

Task: “put a hot potato in fridge” Base model retrieves (ATOMIC/Base): <i>“Apply Task-Specific Physics Reasoning — Heat transfers require closed appliances. Objects in containers remain inaccessible until explicit retrieval.”</i> → Agent focuses on heating procedure but cannot locate the potato → Timeout at 50 steps InsightEmb retrieves (ATOMIC/InsightEmb): <i>“Prioritize Task-Relevant Locations First — Always check object containers/destinations mentioned in the task (e.g., stoveburner, sidetable).”</i> → Agent locates potato → heat → place in fridge → Success in 37 steps
--

Figure 8: Qualitative comparison on Game 0 (in-distribution, top-5 retrieval). The base model’s insight describes the heating *procedure* but not *where to find* the potato; InsightEmb retrieves a search-priority rule that addresses the actual bottleneck.

classify retrieved rules into functional categories—WHERE (search strategy), HOW (state-change procedures), PLACE (destination logic), and VERIFY (state checks)—with GPT-5.2.

The Base model retrieves rules about *how to heat*—“Apply Task-Specific Physics Reasoning,” “Apply Thermodynamic Context Filtering,” “Appliance Function Mapping”—which describe the heating *procedure* but not *where to find the potato*. InsightEmb retrieves rules about *where to search*—

“Prioritize Task-Relevant Locations First,” “Prioritize Direct Affordances,” “Prioritize Immediate Target Interaction”—which address the actual bottleneck: the agent’s first action must be to *locate* the target object.

This reveals a key difference in how the two models understand task structure:

- **Base** matches the query to rules that share the task’s *topic* (heating $\rightarrow$ heating rules). This is *topical matching*.
- **InsightEmb** matches the query to rules that address the task’s *current bottleneck* (the agent hasn’t found the object yet $\rightarrow$ search rules). This is *action-intent matching*: retrieving the rule implied by what the agent needs to do next.

The action-intent match is correct: in ALFWorld, the agent must first *find* the target object before it can apply any state transformation. InsightEmb has learned this sequential dependency from the structural parallel in math, where a problem must first be *understood* (matched to the right strategy) before it can be *solved* (executed step by step).

G ALFWorld Qualitative Case Studies

We present two additional examples where InsightEmb succeeds while Base fails (top-1 retrieval; see Table 11).

Game 9: “clean some ladle and put it in countertop.” InsightEmb checks cabinet 1–3, then sinkbasin 1, opens cabinet 4, finds the ladle on countertop 1, cleans it at the sink, and places it on the countertop—completing in **12 steps**. Base loops through cabinets and the sink without taking the ladle, timing out at 50 steps. This illustrates task-focused search: InsightEmb reaches the sink and visible surfaces early, while Base’s generic “check fridges first” heuristic delays locating the ladle.

Game 33: “put a cool lettuce in countertop.” InsightEmb opens the fridge, scans countertop 1–3, takes the lettuce from countertop 3, cools it in the fridge, and places it on the countertop—completing in **10 steps**. Base opens the fridge but then wanders through drawers and cabinets, never retrieving the lettuce, timing out at 50 steps. Base’s retrieved summary rule prioritizes “common storage areas” broadly; InsightEmb’s rule set emphasizes checking task-mentioned surfaces (countertops) before deep cabinet search.

H Hyperparameters

Hyperparameter	
Base model	Qwen3-Emb-4B
Learning rate	1×10^{-5}
LR scheduler	Cosine
Warmup ratio	0.1
Batch size (per device)	8
Grad. accumulation	2
Training epochs	6
Temperature τ	0.01
Training group size	11
Max passage length	1,536
Precision	BF16
GPUs	8
DeepSpeed	ZeRO-3
Seed	3407

Table 13: Training hyperparameters for both stages.

I Step-Conditioned Retrieval Details

The rule-level analysis above examines *which* rules are retrieved; we now examine *when* they are retrieved. If InsightEmb truly understands procedural structure, its retrieval should *change* as the task progresses: early steps should retrieve search rules (the agent hasn’t found the object yet), while later steps should shift toward state-transformation or verification rules.

We extract the retrieved insight from every step of all 140 in-distribution games (dynamic retrieval with ATOMIC corpus), classify each insight using GPT-5.2 into the same 8 situation labels, and aggregate by task phase. InsightEmb shows phase-aware retrieval: at steps 0–4, it retrieves Search & Locate rules 65.6% of the time; as the task progresses, Verification rules rise from 17.4% to 32.9% and State Transform from 2.6% to 7.5%. In contrast, Base retrieval is phase-invariant: Search & Locate varies only from 18.5% to 27.5%, and State Transform stays between 20.5% and 25.2% across all phases.

Figure 9 shows the per-step stacked area chart (smoothed with a 3-step rolling average).

J WebShop Qualitative Examples

We present illustrative examples from the WebShop evaluation where InsightEmb with ATOMIC insights succeeds while Base fails, or achieves substantially higher task scores. These examples demonstrate the three qualitative patterns identified in §5.2: variant selection awareness, loop prevention, and procedural sequencing.

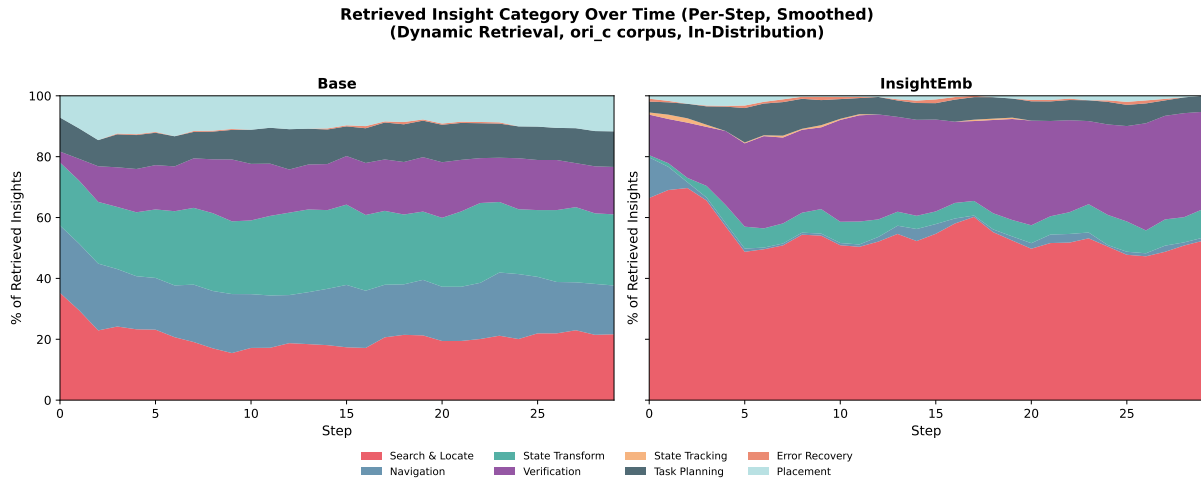


Figure 9: Per-step retrieved insight category distribution (smoothed). InsightEmb shows a clear Search → Verification transition, while Base remains roughly uniform throughout.

J.1 Example 1: Office Chair (Game 226, Atomic)

Task: “Find me height adjustable, high density, easy install, easy assemble home office chairs for living room with color: type 7-pink, and price lower than 120.00 dollars.”

InsightEmb: WON in 6 steps (score = 1.0).

Base: LOST in 3 steps (score = 0.857).

InsightEmb insight:

Progressive Filter Strategy — When initial search yields partial matches, systematically add missing requirements to subsequent queries.

Base insight:

Precision-First Search — Always include all critical attributes in the initial search query using AND logic for better filtering.

Action comparison. Both models find the same product (b08p81rfz4), but:

- **InsightEmb:** search → click product → check features → go back → **select “type 7-pink”** → buy now ✓
- **Base:** search (with AND syntax) → click product → **buy now immediately** (without selecting color variant) ✗

Analysis. InsightEmb’s “Progressive Filter Strategy” guides the agent to inspect product features and select the correct variant before purchasing. Base’s “Precision-First Search” focuses on query formulation but does not instruct variant selection, causing premature purchase with a partial score.

J.2 Example 2: Women’s Sweater (Game 479, Atomic)

Task: “Find me women’s sweaters with relaxed fit, long sleeve with color: light heather grey, and size: large, and price lower than 50.00 dollars.”

InsightEmb: WON in 8 steps (score = 1.0).

Base: LOST in 3 steps (score = 0.600).

InsightEmb insight:

Attribute Cross-Checking — Before any click, verify that the product matches all specified attributes (price, color, size, etc.).

Action comparison. Both find the same product (b07dkgjr74), but:

- **InsightEmb:** search → explore first result → back to search → re-search → click product → **select “light heather grey”** → **select “large”** → buy now ✓
- **Base:** search → click product → **buy now immediately** (without selecting color or size) ✗

Analysis. This is the clearest example of the *variant selection awareness* pattern. InsightEmb’s “Attribute Cross-Checking” rule explicitly guides the agent to verify and select each product variant before purchasing. Base’s generic search-formulation rule does not encode this procedural constraint, and the agent buys immediately without selecting the required color and size variants.

J.3 Example 3: Dining Set (Game 287, Atomic)

Task: “Find me button tufted, mid century, high density, easy assemble dining sets with solid wood,

wood frame for dining room with color: light grey, and price lower than 250.00 dollars.”

InsightEmb: WON in 4 steps (score = 1.0).

Base: LOST in 50 steps (score = 0.000).

This is the most dramatic example ($\Delta = +1.000$).

InsightEmb insight:

Precision-First Search — Always include all critical attributes (type, color, size, price limit) in the initial search query using AND logic for better filtering.

Action comparison.

- **InsightEmb:** search (comprehensive query) → click product (b09gy58gdh) → **select “light grey”** → buy now ✓ (4 steps)
- **Base:** search → back to search → re-search → back to search → re-search → click next → click next → back to search → ... (50 steps, never clicks a product to purchase)

Analysis. InsightEmb’s concise insight produces an effective search query that finds the product immediately. Base gets stuck in a search–refine–next–back loop for all 50 steps, illustrating the *loop prevention* failure: without a clear recovery strategy, the agent exhaustively re-searches without ever committing to a product.

J.4 Example 4: iPad Case (Game 190, Bundle)

Task: “Find me compatible apple online game services with case cover with color: coast coconut trees, and price lower than 50.00 dollars.”

InsightEmb: WON in 6 steps (score = 1.0).

Base: LOST in 50 steps (score = 0.000).

Action comparison.

- **InsightEmb:** search “apple game case cover” → back to search → **re-search “iPad case coconut tree design”** → click product (b07vvvdrcx) → **select “coast coconut trees”** → buy now ✓ (6 steps)
- **Base:** search “Apple case cover” → click next → next → next → next → back to search → re-search → next → next → next → ... (50 steps, navigating pages without clicking any product)

Analysis. InsightEmb’s agent reformulates the search query after the first attempt fails, switching from the literal task wording to a more product-oriented query (“iPad case coconut tree design”). Base’s agent repeatedly paginates through results

without reformulating, illustrating the *loop prevention* pattern: InsightEmb’s retrieval surfaces strategies for query refinement and error recovery.

J.5 Example 5: Men’s Sleepwear (Game 166, Atomic)

Task: “Find me men’s sleep & lounge with long sleeve, elastic waistband for daily wear with color: multi 10, and size: xx-large, and price lower than 80.00 dollars.”

InsightEmb: WON in 9 steps (score = 1.0).

Base: LOST in 7 steps (score = 0.667).

InsightEmb insight:

Compound Feature Recognition — For combined requirements, search for the most distinctive pairing first, then check secondary attributes in product details.

Action comparison.

- **InsightEmb:** search → click product (b09nd9dp7j) → check description → back to search → re-search with exact attributes → click same product → **select “multi 10”** → **select “xx-large”** → buy now ✓
- **Base:** search → click product (b09nd8p2qr) → check features → back to search → re-search → click same product → **buy now without selecting variants** ✗

Analysis. InsightEmb’s “Compound Feature Recognition” guides the agent through a verify–then–select workflow: first confirm the product matches, then explicitly select each variant before purchase. Base’s agent finds a similar product but purchases without selecting the required color and size, resulting in a partial score. This combines *variant selection awareness* with *procedural sequencing*: InsightEmb encodes a sequential workflow (search → verify → select variants → buy).

J.6 Summary of WebShop Qualitative Patterns

Across all divergent games, three consistent patterns emerge:

1. **Variant selection awareness** (Games 226, 479, 393, 166): InsightEmb consistently guides the agent to select product variants (color, size) before purchasing. Base frequently skips this step, resulting in partial scores (0.600–0.857) instead of perfect scores. This is the single most impactful behavioral difference.
2. **Loop prevention** (Games 287, 190): Base gets stuck in search–browse–back loops for 50 steps,

Metric	InsightEmb	Base
Wins (other loses)	9	5
Zero-score games	263	277
Perfect-score games	13	9
Avg task score	0.321	0.282

Table 14: Head-to-head comparison between InsightEmb and Base with ATOMIC insights on 500 WebShop test games.

scoring zero. InsightEmb’s insights about query reformulation and error recovery help the agent find the product and complete the purchase.

3. **Procedural sequencing** (Game 166): InsightEmb retrieves insights that encode a sequential workflow (search $\rightarrow$ verify $\rightarrow$ select variants $\rightarrow$ buy), while Base retrieves topically relevant but procedurally vague insights. This mirrors the ALFWorld finding where InsightEmb performs *procedural matching* rather than *topical matching*.